

Step 1: Initialize the Centroids

We start by picking two random points to act as our "starting averages."

- **Centroid 1 (C1):** (1, 2) — Representing the low-visit, low-spend group.
- **Centroid 2 (C2):** (8, 9) — Representing the high-visit, high-spend group.

Step 2: Distance Calculation (Iteration)

We calculate how far each customer is from C1 and C2 using the Euclidean distance formula:

$$d = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$

Point	x (Visits)	y (Spend)	Distance to C1 (1, 2)	Distance to C2 (8, 9)	Assigned Cluster
P1	1	2	0.00	9.90	1
P2	2	1	1.41	10.00	1
P3	2	4	2.24	7.81	1
P4	1	3	1.00	9.22	1
P5	5	8	7.21	3.16	2
P6	6	9	8.60	2.00	2
P7	7	8	8.49	1.41	2

Point	x (Visits)	y (Spend)	Distance to C1 (1, 2)	Distance to C2 (8, 9)	Assigned Cluster
P8	8	9	9.90	0.00	2
P9	2	2	1.00	9.22	1
P10	7	7	7.81	2.24	2

Step 3: Update the Centroids

Now we find the "true center" of our new groups by averaging the coordinates of every point in that cluster.

For Cluster 1 (P1, P2, P3, P4, P9):

- **New x:** $(1+2+2+1+2) / 5 = 1.6$
- **New y:** $(2+1+4+3+2) / 5 = 2.4$
- *New C1 = (1.6, 2.4)*

For Cluster 2 (P5, P6, P7, P8, P10):

- **New x:** $(5+6+7+8+7) / 5 = 6.6$
- **New y:** $(8+9+8+9+7) / 5 = 8.2$
- *New C2 = (6.6, 8.2)*

Step 4: Final Verification (Iteration 2)

We re-calculate the distances using the **updated** centers. If no customer switches groups, we are done!

Point	x	y	Distance to C1 (1.6, 2.4)	Distance to C2 (6.6, 8.2)	Final Cluster
P1	1	2	0.72	8.35	1
P2	2	1	1.46	8.61	1
P3	2	4	1.65	6.22	1
P4	1	3	0.85	7.64	1
P5	5	8	6.54	1.61	2
P6	6	9	8.04	1.00	2
P7	7	8	7.78	0.45	2
P8	8	9	9.24	1.61	2
P9	2	2	0.57	7.81	1
P10	7	7	7.15	1.26	2